

FINESST Ancillary Documents

A. Facilities, Equipment, and Resources Statement

The proposed work requires modest, presently available resources. There is no need for facility development or shared instrument time. The FI has easy access to all of the resources listed below, so no resource-support letter is required.

Computing. The FI has a workstation with a single consumer-grade GPU (NVIDIA GeForce GTX 1070 Ti, 8 GB of VRAM) and at least 2 TB of local storage, owned by the research group and the FI. This is more than sufficient for all model training and elevation processing in the proposal.

Workspace. The FI has office and lab space in NCALM (National Center for Airborne Laser Mapping) at the University of Houston, with institutional backup storage for project data.

Software. The full processing stack is open source and requires no subscriptions or commercial licenses (Python scientific stack, PDAL, GDAL/OGR, WhiteboxTools, PyTorch, QGIS).

Field work. None is proposed. The project is based entirely on existing datasets and requires no Antarctic deployment, logistics support, or permitting.

B. Acknowledgements

The Future Investigator (FI) is the primary author of this proposal. The FI thanks Dr. Craig Glennie for his scientific mentorship and guidance in shaping and editing the research, and Cami Barlow, whose dissertation on Antarctic stream-channel detection and geomorphic change analysis provides the foundation this project builds upon. AI assistance (Anthropic's Claude, used as an editing tool under the FI's direction) contributed to light grammar and structure editing, document formatting, and data-availability verification; all scientific content, claims, and decisions are the FI's own and were reviewed by the FI and PI. Publicly archived datasets are credited to their providers: NASA ATM (via USGS), NCALM/OpenTopography, the Polar Geospatial Center (REMA), the McMurdo Dry Valleys LTER (via the Environmental Data Initiative), Copernicus/ECMWF (ERA5), and NCAR (AMPS).